

Note on higher-form symmetries and anomalies

Ruihua Fan¹

¹*Department of Physics, Harvard University, Cambridge MA 02138, USA*

Abstract

This note discusses the higher form symmetries and their anomalies. We focus on the 0-form and 1-form symmetries and study them in concrete examples.

Contents

1	Introduction	1
2	Higher-form symmetry: kinematics	2
3	General comments on 't Hooft anomaly	3
4	Toy models	4
4.1	2d Maxwell theory with θ angle	5
4.2	3d Maxwell theory	6
5	Interacting cases	7
5.1	2d abelian Higgs model	7
5.2	3d abelian Higgs model	8

1 Introduction

Global symmetries play important roles in all physics studies. They not only provide constraints on kinematics but can also have strong implications on dynamics. Therefore it crucial to have a better understanding on the global symmetry. Recently, the notion of global symmetries is generalized by Ref. [1]. The original symmetries, where the charged objects live on points, are generalized to q -form symmetries, where the charged objects live on q -dimensional closed manifolds. This generalization allows us to understand originally distinct phenomena in a unified framework and has inspired many discussion.

In this note, we will explore the 't Hooft (mixed) anomaly of higher-form symmetries and its implication on the dynamics. In Sec.2, we give a brief summary of the properties of higher-form symmetries and focus on its kinematic aspects. In Sec.3, we discuss how the 't Hooft anomaly can constrain the dynamics, especially the IR properties. In Sec.4, we discuss two toy models where we can solve the model exactly and compare with the result of computing anomaly. This prepares us for the discussion in Sec.5, where we generalize the toy models in Sec.4 to interacting cases, which are and no longer exactly solvable. But we can still use anomaly to make predictions on the IR properties.

2 Higher-form symmetry: kinematics

In this section, we give a brief summary of the properties of q -form (unitary) symmetries. Let us consider the d -dimensional spacetime, then a q -form (unitary) symmetry G is characterized by the followings

1. Charged objects live on q -dimensional closed manifolds, denoted by $V(C^{(q)})$
2. Symmetry action is associated to a topological surface operator supported on $(d - q - 1)$ -dimensional closed manifolds, denoted by $U_g(M^{(d-q-1)})$. The group law is dictated by

$$U_g(M^{(d-q-1)})U_{g'}(M^{(d-q-1)}) = U_{g''}(M^{(d-q-1)}), \quad g'' = gg' \quad (1)$$

3. The operator $U_g(M^{(d-q-1)})$ has non-trivial action on the charged object $V(C^{(q)})$ when there is linking between $M^{(d-q-1)}$ and $C^{(d)}$, i.e.

$$U_g(M^{(d-q-1)})V(C^{(q)}) = R_V(g)V(C^{(q)}) \quad (2)$$

where $R_V(g)$ is the representation of $g \in G$ carried by the operator V . For a 0-form symmetry and $U_g(\Sigma^{(d-1)})$ supported on a constant time slice $\Sigma^{(d-1)}$, we can take a charged object $O_i(p)$ lives at a single point p and have

$$U_g(\Sigma^{(d-1)})O_i(p)U_g(\Sigma^{(d-1)}) = R(g)_i^j O_j(p), \quad p \in \Sigma^{(d-1)} \quad (3)$$

where $R(g)$ is the representation of the group element $g \in G$.

4. The topological nature of U_g and the symmetry action implies that $(q > 0)$ -form symmetries must be abelian. For example, let two operators U_{g_1} and U_{g_2} link with $V(C^{(q>0)})$, then there is no relative ordering between U_{g_1} and U_{g_2} .¹
5. We can couple the system to a $(q+1)$ -form gauge field. Doing this for a discrete symmetry in the continuum might be mathematically complicated. On the lattice, one can simply attach gauge degrees of freedom to the plaquette for 1-form symmetries or higher-dimensional simplices for higher-form symmetries.

When the symmetry is continuous, we can follow the Noether procedure to define a current

1. The conserved current J is a $(q+1)$ -form. It is equivalent to use its Hodge dual $j = \star J$,² which is a $(d - q - 1)$ -form. The conservation law is

$$\partial_\mu J^{\mu\nu_2 \dots \nu_d} = 0 \Leftrightarrow d \star J = 0 \quad (5)$$

2. There are two charge operators one can define. One is defined as the integral of current $j = \star J$ on a closed $(d - q - 1)$ -dimension manifold

$$Q(M^{(d-q-1)}) = \int_{M^{(d-q-1)}} j \quad (6)$$

¹We thank Xi Yin for pointing this out.

²Our convention of Hodge dual follows Nakahara and is

$$\star J = \frac{1}{(d-q)!} J_{\mu_1 \dots \mu_q} \epsilon^{\mu_1 \dots \mu_q}_{\nu_{q+1} \dots \nu_d} dx^{\nu_{q+1}} \wedge \dots \wedge dx^{\nu_d} \quad (4)$$

The convention of the totally anti-symmetric tensor is defined as $\epsilon_{012 \dots d} = +1$ and one raises the index by contracting with the metric tensor.

The symmetry action is related by an exponential

$$U_g(M^{(d-q-1)}) = e^{i\alpha_g Q(M^{(d-q-1)})} \quad (7)$$

The other is the standard one, which is defined on a constant time slice

$$Q_\lambda = \int_{space} \lambda \wedge j \quad (8)$$

where λ is closed q -form so that Q_λ is independent of time. If we embed $M^{(d-q-1)}$ into some constant time slice, $Q(M^{(d-q-1)})$ will coincide with Q_λ if one choose λ to be the Poincaré dual of $M^{(d-q-1)}$ in the constant time slice (not the Poincaré dual in the whole spacetime).³ In the following discussion, we will only use $Q(M^{(d-q-1)})$. The second definition will be convenient in the discussion of spontaneous symmetry breaking and Goldstone's theorem, which is not the focus of this note.

3. Gauging a continuous q -form symmetry is relatively easy, we need to add to the Lagrangian

$$\Delta L \sim B_{\mu_1 \dots \mu_q} J^{\mu_1 \dots \mu_q} \sim B \wedge \star J \quad (11)$$

The reason we use $\sim$ instead of $=$ is that sometimes one might also need to add terms only involving B itself (e.g. the Abelian Higgs model).

3 General comments on 't Hooft anomaly

A global symmetry G has an 't Hooft anomaly when it cannot be gauged. It has far-reaching consequences on the dynamics of the quantum system. We give a short review in this section.

We first give a concrete definition of 't Hooft anomaly [2]. Let $Z[A]$ be the partition function with a background gauge field A of the G . Then G has an 't Hooft anomaly if $Z[A]$ has a intrinsic phase ambiguity under a gauge transformation $A \rightarrow A + \alpha$

$$Z[A + \alpha] = e^{i\phi[\alpha, A]} Z[A], \quad \phi[\alpha, A] = \int d^d x \mathcal{L}_{anomaly}[\alpha, A] \quad (12)$$

where “intrinsic” means that the phase ambiguity cannot be absorbed by any d -dimensional local counter-terms of A , and $\mathcal{L}_{anomaly}[\alpha, A]$ is called the anomaly polynomial.

We have not specified the precise meaning of a local counter-terms of A . Now, we will give a physical definition. A mathematically rigorous definition is way beyond the scope of this note. Suppose the system has a unique gapped ground state on a topologically trivial closed manifold. We turn on a background gauge field A and compute the partition function

$$Z[A] = e^{iS_{eff}[A]} = \exp(i \int d^d x \mathcal{L}_{local}[A]) \quad (13)$$

³Let us give one concrete example. We consider $3d$ spacetime and a 1-form symmetry, so the closed current j is also a 1-form. Let the constant time slice be the xy -plane and $M^{(1)}$ be the x -axis, i.e.,

$$Q(M^{(1)}) = \int_{M^{(1)}} j = \int_{y=0} dx j_x \quad (9)$$

We can choose a 1-form $\lambda = -\delta(y)dy$, which is closed by construction. One can check that

$$Q_\lambda = \int_{\mathbb{R}^2} \lambda \wedge j = \int \lambda_\mu j_\nu dx^\mu \wedge dx^\nu = \int_{y=0} dx j_x \quad (10)$$

Local counter-terms can be regarded as the possible effective actions $S_{eff}[A]$ obtained in this procedure, and 't Hooft anomaly means $\delta_\alpha \mathcal{L}_{local}[A] \neq \mathcal{L}_{anomaly}[\alpha, A]$ for any $\mathcal{L}_{local}$.⁴ In condensed matter systems, there are always unavoidable trivially gapped degrees of freedom, the existence of such local counter-terms is actually physical.

Now, let us understand the physical consequences of 't Hooft anomaly. It follows from the above definition that having 't Hooft anomaly of a global symmetry G implies the IR cannot be a unique gapped vacuum state. Instead, it can have⁵

1. gapless without breaking the symmetry G
2. more than one vacuum state, which either comes from being a topological field theory or spontaneously symmetry breaking of G .

One can see that the above statement is independent of what the global symmetry is and thus is also applicable to higher-form symmetries. Moreover, the above statement is about the IR behavior, thus it is sufficient for the involved symmetry to emergent.

In many situations, the symmetry group factorizes $G = G_1 \times G_2$ (we assume it is direct product for simplicity) and we call this type of anomaly a mixed 't Hooft anomaly. Famous examples are the chiral anomalies found in 2d and 4d free Dirac fermions [3] and the parity anomaly in 3d free Dirac fermions. When this is the case, adding the background field of one symmetry breaks the other explicitly. For simplicity, let us assume G_1, G_2 are continuous symmetries and add the background field A_1 for the symmetry G_1 to show the current conservation of G_2 breaks down. We can perform the Noether procedure of G_2 , which amounts to turning on an infinitesimal gauge field $A_2 = d\alpha_2$. Using the assumption of 't Hooft anomaly, we have

$$Z[A_1, d\alpha_2] - Z[A_1, 0] = \int D\varphi (\star j_2 \wedge d\alpha_2 + \alpha_2 f[A_1]) e^{iS[\varphi, A_1]} + \mathcal{O}(\alpha_2^2) \quad (14)$$

which directly show that the current $\star j_2$ is no longer conserved.

The main value in thinking about 't Hooft anomaly is that it allows one to make non-perturbative (and exact) statements about renormalization group flows that cannot be understood using perturbation theory.⁶ In this note, we will not prove this statement but simply try to apply it in some concrete examples.

However, anomaly are usually hard to compute. Some can be identified at the one-loop order while others may require harder work. The discussion in this note takes the following strategy. We will deform the full (interacting) theory in a symmetry-preserving way towards some limits, where the anomaly becomes computable (actually manifest even at the tree level). The results obtained at these limits will help us to make predictions at the intermediate coupling.

4 Toy models

In the following, we will discuss some toy models that can be viewed as certain limit of the fully interacting models. These models are free so that one can compute the anomaly and check the results exactly. The anomalies we will see are both a mixed anomaly between a 0-form and a 1-form symmetry. Although the symmetries in both examples are accidental (relying on the fine tuning), they can be emergent in the full theories.

⁴For a continuous symmetry, A is an ordinary smooth function. Then one can imagine $\mathcal{L}_{local}$ to be some function that only depends on derivatives of A . The locality is due to the fact that and the higher order terms are suppressed by the energy gap (mass). For a discrete symmetry, what quantifies as a local counter-term is a complicated question.

⁵We thank Minjae Cho for helpful discussion.

⁶We thank Xi Yin for this comment.

4.1 2d Maxwell theory with θ angle

The first model is the 2d Maxwell theory with a θ angle

$$\mathcal{L} = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} - \frac{\theta}{2\pi} da \quad (15)$$

where $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ is the field strength and a is a $U(1)$ gauge field. By the Dirac quantization condition $\int_{M^{(2)}} da \in 2\pi\mathbb{Z}$, therefore θ is 2π periodic. Depending on the value of θ , the system can have either one or two global symmetries:

1. 1-form symmetry $U(1)^e$, whose charge operator is

$$Q^{(1)} = \frac{1}{e^2} \star f \quad (16)$$

Its conservation is due to the equation of motion $d \star f = 0$. The corresponding charged objects are the Wilson loops

$$W(C^{(1)}) = \exp \left(i \oint_{C^{(1)}} a \right) \quad (17)$$

2. 0-form symmetry, charge conjugation, that only appears at $\theta = 0, \pi$

$$C : a \rightarrow -a \quad (18)$$

Our claim is that there is a mixed anomaly between the charge conjugation and 1-form symmetry at $\theta = \pi$. To show this, let us turn on a background gauge field B for the 1-form symmetry

$$\mathcal{L}[a] \rightarrow \mathcal{L}[a, B] = -\frac{1}{4e^2} (da - B) \wedge \star (da - B) - \frac{\theta}{2\pi} (da - B) \quad (19)$$

Now, under a charge conjugation, the action is changed by (mod 2π)

$$\frac{\theta}{2\pi} \int_{M^{(2)}} B \quad (20)$$

which can be arbitrary value. It was shown that there is no 2d counter-term that can cancel the above variation [4], which proves the mixed anomaly at $\theta = \pi$.

Now, let us understand the implication of this anomaly on the IR dynamics. On one hand, 1-form symmetries cannot be spontaneously broken in 2d [1]. On the other hand, the theory defined on a line is gapped due the lack of propagating mode. We know that the IR must either spontaneously break the charge conjugation or be a TQFT. It is the first case that happens here.

The easiest way of showing this is to actually solve the model by placing the theory on a circle of radius R . Then the effective action of a (in the gauge $a_0 = 0$) is

$$\mathcal{L} = \frac{1}{2e^2 R} \dot{q}^2 + \frac{\theta}{2\pi} \dot{q}, \quad q = \int_{S^1} a \quad (21)$$

which shows a two-fold degenerate ground states at $\theta = \pi$ with the wavefunction being $e^{\pm i q}$. One can see that the two ground states are related by the charge conjugation. Thus the IR spontaneously breaks the charge conjugation.

4.2 3d Maxwell theory

The second model is the 3d pure Maxwell theory (without the Chern-Simons term)

$$\mathcal{L} = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} \quad (22)$$

where $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ is the field strength and a is a $U(1)$ gauge field. The field strengths satisfies two equations, one is the Bianchi identity and the other is the equation of motion

$$df = 0, \quad d \star f = 0 \quad (23)$$

These two equations are associated to two global symmetries:

1. 0-form $U(1)$ symmetry, whose charge operator is defined as

$$Q_{MX}^{(0)}(M^{(2)}) = \frac{1}{2\pi} \int_{M^{(2)}} f \quad (24)$$

which, when $M^{(2)}$ is a constant time slice, measures the magnetic flux. The Dirac quantization condition implies that $Q_{MX}^{(0)} \in \mathbb{Z}$. The corresponding conserved current is

$$J^m = -\frac{1}{2\pi} \star f \quad (25)$$

where the minus sign comes from the Lorentzian signature. The charged object is the monopole operator, which cannot be written down by any local functions of f .

2. 1-form $U(1)$ symmetry, whose charge operator is defined as

$$Q_{MX}^{(1)}(M^{(1)}) = \frac{1}{e^2} \int_{M^{(1)}} \star f \quad (26)$$

which, when $M^{(1)} \in \Sigma^{(2)}$ is in a constant time slice, measures the electric charge. Having a $U(1)$ gauge field implies $Q_{MX}^{(1)} \in \mathbb{Z}$. The corresponding conserved current is

$$J^e = -\frac{1}{e^2} f \quad (27)$$

and the charged objects are the Wilson line operators

$$W(C^{(1)}) = \exp \left(in \oint_{C^{(1)}} a \right), n \in \mathbb{Z} \quad (28)$$

Our claim is that there is a mixed anomaly between $U(1)^m$ and $U(1)^e$. To detect it, we add background field for the 0-form symmetry, i.e.

$$\mathcal{L}[a] \rightarrow \mathcal{L}[a, A] = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} + A_\mu J^{m,\mu} = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} - \frac{1}{2\pi} \epsilon^{\mu\nu\rho} A_\mu \partial_\nu a_\rho \quad (29)$$

Then the equation of motion of f is modified to

$$\partial_\mu f^{\mu\nu} = -\frac{e^2}{2\pi} \epsilon^{\rho\sigma\nu} \partial_\rho A_\sigma \quad (30)$$

which implies the explicitly breaking of $U(1)^e$. Notice that A is not a gauge invariant quantity, thus the right-hand side, despite being a total derivative, cannot be simply absorbed by a redefinition of f .⁷ Therefore, this is a manifestation of a mixed 't Hooft anomaly.

Now, let us understand the implication of this anomaly on the IR dynamics. One might naively say that the theory is gapless and consistent with the general constraints. But we have to be careful. There are two possibilities. One is that the theory is symmetric gapless. The other is that the theory spontaneously break one symmetry⁸ and gaplessness comes from the Goldstone mode. It is the second case that happens here.

The easiest way of proving this is to use the particle-vortex duality. Namely, the 3d Maxwell theory is secretly identical to the 3d compact scalar field theory described by

$$\mathcal{L}_S[\phi] = -\frac{\kappa}{2}(\partial\phi)^2 \quad (31)$$

The fundamental fields and global symmetries are identified as follows

$$\begin{aligned} d\phi &\leftrightarrow \star f \\ \frac{1}{2\pi} \int f &\leftrightarrow 2\kappa \int \star d\phi \\ \frac{1}{e^2} \int \star f &\leftrightarrow \frac{1}{2\pi} \int d\phi \end{aligned} \quad (32)$$

Consequently, the monopole operator has a simple expression in terms of ϕ

$$\mathcal{M} \leftrightarrow e^{i\phi} \quad (33)$$

$\mathcal{L}_S[\phi]$ describes a superfluid that spontaneously breaks the 0-form $U(1)$ symmetry, namely the equal-time two point function $\langle e^{i\phi(x)} e^{-i\phi(0)} \rangle$ approaches a non-zero value when $|x| \rightarrow 0$. It implies the spontaneous breaking of $U(1)^m$ of the 3d Maxwell theory.

5 Interacting cases

5.1 2d abelian Higgs model

In this section, we generalize the 2d Maxwell theory to the 2d abelian Higgs model by including a single charge-1 matter field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} - \frac{\theta}{2\pi} da - |D\varphi|^2 - m^2|\varphi|^2 - \lambda|\varphi|^4 \quad (34)$$

The charge conjugation symmetry is still present at $\theta = \pi$, while the matter field breaks the 1-form symmetry $U(1)^e$ explicitly. This model cannot be solved exactly, but we can use anomaly to make many useful predictions. In the following discussion, we will put the theory on a plane.

Let us analyze the theory in certain limits:

- First, let us consider the limit $m^2 \gg e^2$. In this case, the matter field is gapped and can be integrated out, which will generate terms like f^n that are suppressed by the mass and lead to a local effective action of a . These terms will correct the kinetic term of the gauge field, however, we can still define a conserved current with its exact form modified by the quantum

⁷We thank Xi Yin for this comment.

⁸in this case, it has to be the 0-form symmetry due to the generalized Coleman-Mermin-Wagner theorem [1].

corrections from the matter field. This shows that the 1-form symmetry although absent in the UV is emergent in the IR. Then we can repeat the anomaly analysis in Sec. 4 and claim a unique ground state for $\theta \neq \pi$ and a two-fold ground state degeneracy at $\theta = \pi$. Namely, in the limit $m^2 \gg e^2, \lambda$, we can have a first order phase transition by tuning θ , with $\theta = \pi$ being the transition point. This behavior is reminiscent to the 4d pure Yang-Mills theory [5].

- Then, we consider the opposite limit $-m^2 \gg e^2$. The negative potential implies that ϕ develops a non-zero expectation value and it is a good approximation to assume the decoupling of the radial mode of ϕ . Then the gauge field is Higgsed by absorbing the angular mode of ϕ , which implies a unique charge-conjugation symmetric gapped ground state for all θ .
- Let us focus on $\theta = \pi$ and tune the mass m^2 . The system is gapped and spontaneously breaks the charge conjugation at $m^2 \gg e^2$ and is gapped symmetric at $-m^2 \gg e^2$. This scenario is reminiscent to the case of Ising phase transition. Therefore, one possible scenario is that there is a second order phase transition at the intermediate m^2 . However, other possibilities are also possible.

5.2 3d abelian Higgs model

In this section, we generalize the 3d Maxwell theory to the abelian Higgs model by including a single charge-1 matter field. The Lagrangian is given by

$$\mathcal{L} = -\frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} - |D\varphi|^2 - m^2 |\varphi|^2 - \lambda |\varphi|^4 \quad (35)$$

Now the 0-form symmetry $U(1)^m$ is still present but the 1-form symmetry $U(1)^e$ is explicitly broken by the matter field.

Now, we can simply repeat all the analysis above for the 2d abelian Higgs model with $\theta = \pi$. Then we will find that the system spontaneously breaks $U(1)^m$ for $m^2 \gg e^2, \lambda$ and is gapped symmetric at $-m^2 \gg e^2$. Therefore, one possible scenario is that there is a second order phase transition at the intermediate m^2 . However, we cannot rule out other possibilities.

Acknowledgment

We thank Xi Yin for many helpful instructions and answering related questions during the preparation of this topic. We thank Minjae Cho, Liujun Zou for helpful discussions.

References

- [1] D. Gaiotto, A. Kapustin, N. Seiberg and B. Willett, *Generalized Global Symmetries*, *JHEP* **02** (2015) 172 [[1412.5148](#)].
- [2] C. Córdova, T. T. Dumitrescu and K. Intriligator, *Exploring 2-Group Global Symmetries*, *JHEP* **02** (2019) 184 [[1802.04790](#)].
- [3] X. Yin, *Lectures on quantum field theories*, .
- [4] Z. Komargodski, A. Sharon, R. Thorngren and X. Zhou, *Comments on Abelian Higgs Models and Persistent Order*, *SciPost Phys.* **6** (2019) 003 [[1705.04786](#)].
- [5] D. Gaiotto, A. Kapustin, Z. Komargodski and N. Seiberg, *Theta, Time Reversal, and Temperature*, *JHEP* **05** (2017) 091 [[1703.00501](#)].